

MAE 3240: Problem Set #3: Solutions

Name: Nitya Shukla

Date of Submission: 02/24/2026

Problem Set Number: #3

Due Date: 02/24/2026

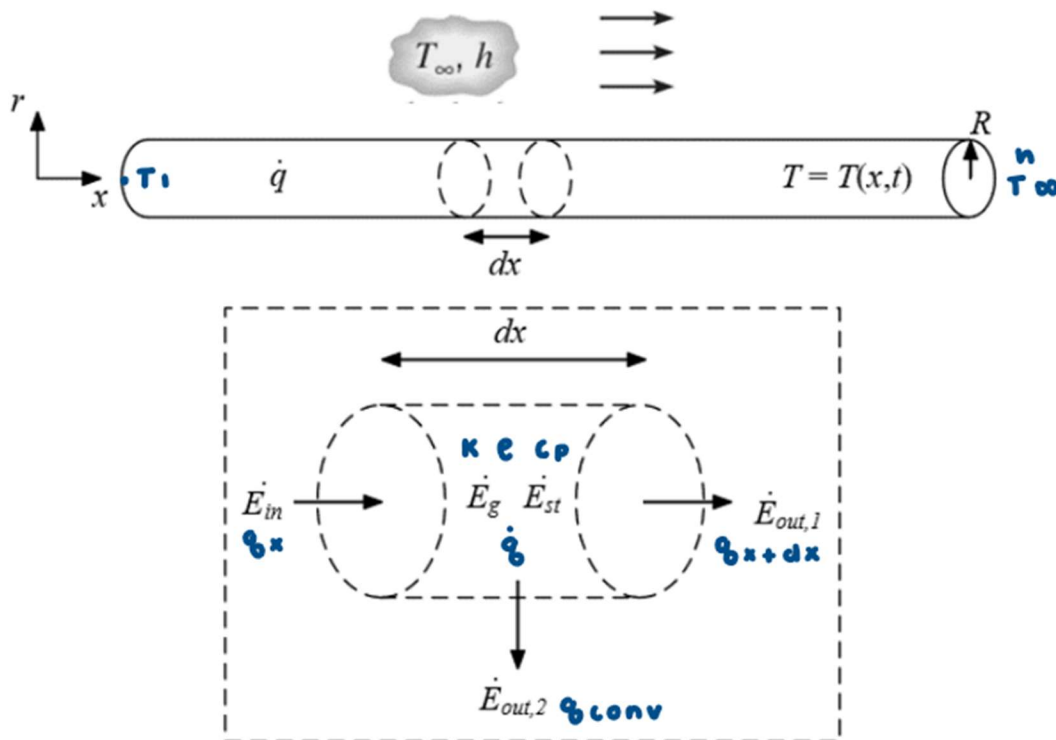
Acknowledgement: Discussed solutions to problem set with Thomas Wells and April Wang.

Problem #1

Known: Radius (R), length (L), volumetric heat generation rate ($\dot{q}$), ambient temperature (T_∞), heat transfer coefficient (h)

Find: (a) Energy balance equation of control volume as function of T , (b) Derivation of governing differential equation that describes temperature profile or rod, (c) Boundary conditions and initial conditions for governing differential equation

Schematic:



Properties: Heat transfer coefficient (h)

Assumptions: (1) Uniform volumetric heat generation rate ($\dot{q}$), (2) Negligible thermal radiation, (3) Radius of rod is small, (4) Temperature of rod along radial direction is uniform, (5)

Temperature of rod is function of x -coordinate and time, i.e., $T = T(x, t)$, (6) Constant thermal conductivity (k), density (ρ), and heat capacity (c_p), (7) Boundary conditions: constant temperature T_1 at $x = 0$, and air flow with temperature T_∞ and heat transfer coefficient h at $x = L$, (8) Initial condition: rod has uniform initial temperature T_i at $t = 0$ s

Analysis:

① (a) Energy Balance: $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} = \dot{E}_{st}$

Fourier's Law: $q_x = -kA \frac{\partial T}{\partial x}$; $q_{conv} = hA_s(T - T_\infty)$

Taylor Series Expansion: $q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx$

$\dot{E}_{in} = q_x = -kA \frac{\partial T}{\partial x} = -k(\pi R^2) \frac{\partial T}{\partial x}$ $\downarrow$
 $= q_x + (-kA) \frac{\partial^2 T}{\partial x^2} dx$

$\dot{E}_g = \dot{q} A dx = \dot{q} (\pi R^2) dx$ $A = \pi R^2$; $A_s = 2\pi R dx$

$\dot{E}_{st} = \rho C_p \frac{\partial T}{\partial t} A dx = \rho C_p \frac{\partial T}{\partial t} (\pi R^2) dx$

$\dot{E}_{out,1} = q_{x+dx} = q_x + (-kA) \frac{\partial^2 T}{\partial x^2} dx = -k\pi R^2 \left[\frac{\partial T}{\partial x} + \frac{\partial^2 T}{\partial x^2} dx \right]$

$\dot{E}_{out,2} = q_{conv} = hA_s(T - T_\infty) = h(2\pi R dx)(T - T_\infty)$

(b) $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} = \dot{E}_{st}$

$q_x + \dot{q} A dx - q_{x+dx} - hA_s(T - T_\infty) = \rho C_p \frac{\partial T}{\partial t} A dx$

$\dot{q} (\pi R^2) dx + k(\pi R^2) \frac{\partial^2 T}{\partial x^2} dx - h(2\pi R)(T - T_\infty) dx = \rho C_p \frac{\partial T}{\partial t} (\pi R^2) dx$

$R^2 \dot{q} + kR^2 \frac{\partial^2 T}{\partial x^2} - 2hR(T - T_\infty) = \rho C_p R^2 \frac{\partial T}{\partial t}$

$kR \frac{\partial^2 T}{\partial x^2} - \rho C_p R \frac{\partial T}{\partial t} - 2hT = -2hT_\infty - R \dot{q}$

(c) Boundary conditions: $x=0 \rightarrow$ Const. Surface temp.

$T(0, t) = T_i, t > 0$

$x=L \rightarrow$ convection $\Rightarrow k \frac{\partial T}{\partial x} \bigg|_{x=L} = h[T_\infty - T(L, t)]$

$\frac{\partial T(L, t)}{\partial x} = \frac{h}{k} [T_\infty - T(L, t)], t > 0$
 $0 \leq x \leq L$

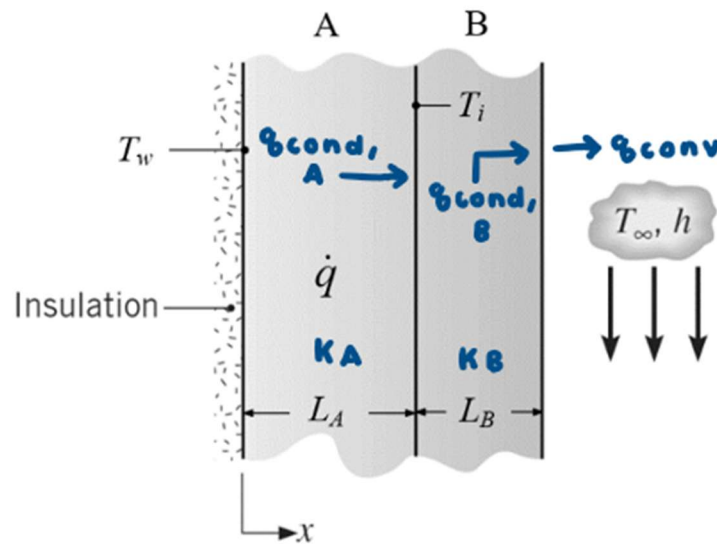
Initial condition: uniform temp. $\Rightarrow T(x, 0) = T_i, 0 \leq x \leq L$

Problem #2

Known: Thickness of wall A (L_A) and wall B (L_B), thermal conductivity of wall B (k_B), coolant temperature (T_∞), coolant convective heat transfer coefficient, temperature between thermal insulation and wall A (T_w), temperature between wall A and B (T_i)

Find: (a) Right surface temperature T_s of wall B, (b) Volumetric heat generation $\dot{q}$ at steady-state, (c) Thermal conductivity k_A at wall A, (d) Plot of temperature profile across wall A and B

Schematic:



Properties: Thermal conductivity of wall B (k_B), coolant convective heat transfer coefficient

Assumptions: (1) One-dimensional heat conduction, (2) Uniform volumetric heat generation rate ($\dot{q}$) at wall A and no heat generation within wall B, (3) Left surface of wall A thermally insulated and right surface of wall B exposed to coolant, (4) No contact thermal resistance at boundary between wall A and B

Analysis:

$$\textcircled{2} \text{ (a) Heat equation: } \frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\text{Steady-State, 1D, } k_B = \text{const, } \dot{q} = 0 \Rightarrow k_B \frac{\partial^2 T}{\partial x^2} = 0 \rightarrow \frac{\partial^2 T}{\partial x^2} = 0$$

$$\frac{\partial T}{\partial x} = \int \frac{\partial^2 T}{\partial x^2} dx = C_1 \rightarrow T = \int \frac{\partial T}{\partial x} dx = C_1 x + C_2$$

Boundary Conditions: $x=0 \rightarrow T(0,t)=T_i, t>0$

$$x=L_B \rightarrow \frac{\partial T(L_B,t)}{\partial x} = \frac{h}{K_B} [T_\infty - T_s], t>0$$
$$C_1 = \frac{h}{K_B} [T_\infty - T_s], C_2 = T_i \quad 0 \leq x \leq L_B$$

$$T(x) = \frac{h}{K_B} [T_\infty - T_s] x + T_i \Rightarrow T_s = \frac{h}{K_B} [T_\infty - T_s] L_B + T_i$$

$$T_s + \frac{h}{K_B} L_B T_s = \frac{h}{K_B} T_\infty L_B + T_i$$

$$T_s = \frac{h}{K_B} T_\infty L_B + T_i = \left(\frac{50}{3} \right) (20 + 273.15) (0.1) + (100 + 273.15)$$

$$\frac{1 + \frac{h}{K_B} L_B}{1 + \left(\frac{50}{3} \right) (0.1)}$$

$$T_s = 323.15 \text{ K} = 50^\circ \text{C}$$

(b) Steady-State: $\dot{E}_g = \dot{E}_{out} \Rightarrow q''_{gen} = q''_{conv}$

$$h(T_s - T_\infty) = \dot{q} L_A \Rightarrow \dot{q} = \frac{h(T_s - T_\infty)}{L_A} = \frac{(500)(50 - 20)}{0.3} = \boxed{50000 \frac{\text{W}}{\text{m}^3}}$$

(c) Heat equation: $\frac{\partial}{\partial x} \left(K_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_z \frac{\partial T}{\partial z} \right) + \dot{q} = \rho C_p \frac{\partial T}{\partial t}$

Steady-State, 1D, $K_A = \text{const} \Rightarrow K_A \frac{\partial^2 T}{\partial x^2} + \dot{q} = 0$

$$\frac{\partial^2 T}{\partial x^2} = -\frac{\dot{q}}{K_A} \rightarrow \frac{\partial T}{\partial x} = \int \frac{\partial^2 T}{\partial x^2} dx = -\frac{\dot{q}}{K_A} x + C_1$$

$$T = \int \frac{\partial T}{\partial x} dx = -\frac{\dot{q}}{2 K_A} x^2 + C_1 x + C_2$$

Boundary Conditions: $x=0 \rightarrow \text{insulation} \Rightarrow \frac{\partial T(0,t)}{\partial x} = 0, t>0$

$$x=0 \rightarrow \text{const. surface temp.} \Rightarrow T(0, t) = T_w$$

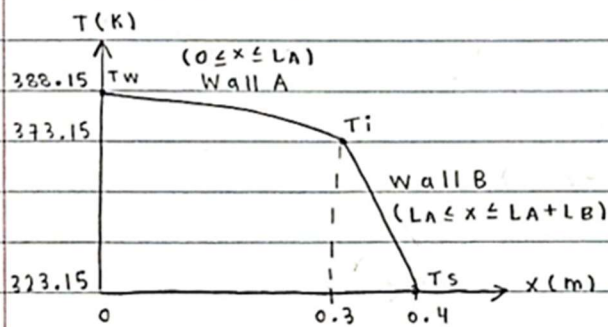
$$C_1 = 0, C_2 = T_w \Rightarrow T(x) = \frac{-\dot{q} x^2}{2k_A} + T_w$$

$$T(L_B) = T_i = \frac{-\dot{q} (L_A)^2}{2k_A} + T_w \Rightarrow k_A = \frac{-\dot{q} (L_A)^2}{2(T_i - T_w)}$$

$$k_A = \frac{-(50000)(0.3)^2}{2(100 - 115)} = 150 \text{ W/(m-K)}$$

$$(d) 0 \leq x \leq L_B: T(x) = \frac{h}{k_B} [T_w - T_s] x + T_i = -500x + 373.15 \text{ K}$$

$$0 \leq x \leq L_A: T(x) = \frac{-\dot{q} x^2}{2k_A} + T_w = \frac{-500 x^2}{3} + 388.15$$

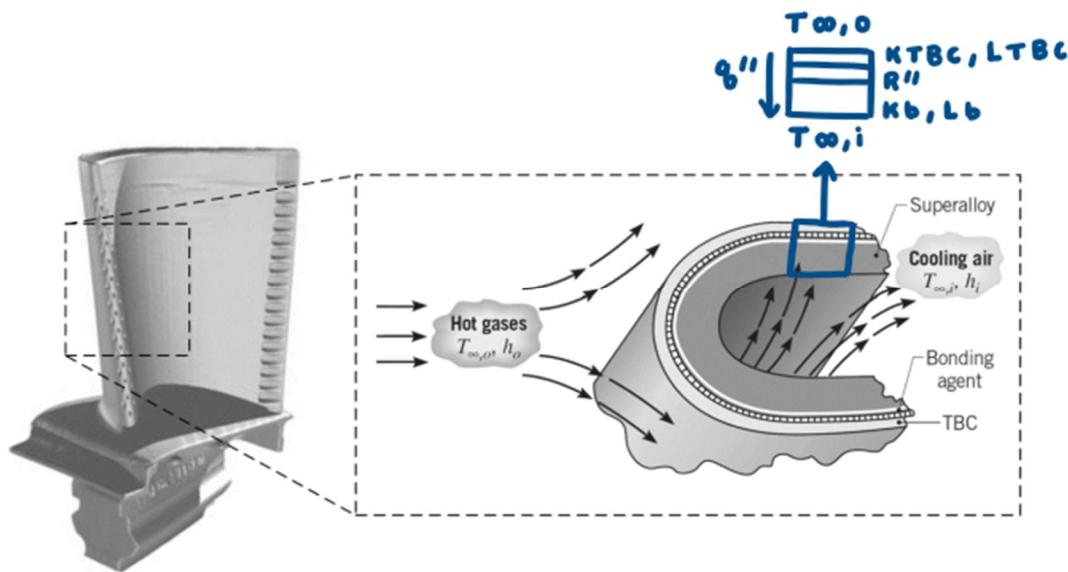


Problem #3

Known: Superalloy blade thermal conductivity (k_b), ceramic TBC thermal conductivity (k_{TBC}), interfacial thermal resistance (R''), temperature of external hot gases ($T_{\infty, o}$) and cooling air inside blade ($T_{\infty, i}$), heat transfer coefficient of hot gases and cooling air (h_o), thickness of blade (L_b) and TBC (L_{TBC}), maximum allowable temperature of blade (T_{max})

Find: (a) (TBC not applied) Resistance network, exterior surface temperature of blade, threshold temperature; (b) (TBC applied) Resistance network, exterior surface temperature of blade, threshold temperature

Schematic:



Properties: Superalloy blade thermal conductivity (k_b), ceramic TBC thermal conductivity (k_{TBC}), heat transfer coefficient of hot gases and cooling air (h_o)

Assumptions: (1) Negligible thickness of metallic bonding agent, (2) Steady-state condition, (3) Negligible thermal radiation, (4) Blade geometry approximated as plane wall: one-dimensional heat conduction

Analysis:



$$R''_{th,1} = R''_{conv,o} + R''_{cond,b} + R''_{conv,i} = \frac{1}{h_o} + \frac{L_b}{K_b} + \frac{1}{h_i}$$

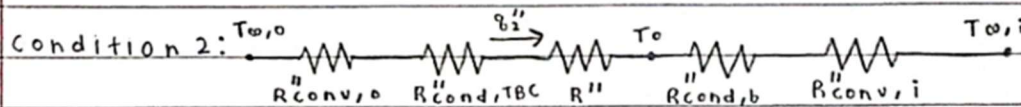
$$R''_{th,1} = \frac{1}{1000} + \frac{(5 \times 10^{-3})}{25} + \frac{1}{500} = 0.0032 \frac{(m^2 \cdot K)}{W}$$

$$q''_1 = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{th,1}} = \frac{1616 - 400}{0.0032} = 380 \frac{KW}{m^2}$$

$$T_{\infty,o} - T_o = q''_1 \Rightarrow T_o = T_{\infty,o} - q''_1 R''_{conv,o} = 1616 - (380 \times 10^3) \left(\frac{1}{1000} \right) = 1236 K$$

Superalloy blade not maintained

under T_{max} ($1236 > 1200 K$)



$$R''_{th,2} = R''_{th,1} + R''_{cond,TBC} + R'' = R''_{th,1} + \frac{L_{TBC}}{K_{TBC}} + R''$$

$$R''_{th,2} = 0.0032 + \frac{(0.5 \times 10^{-3})}{1} + 10^{-4} = 0.0038 \frac{(m^2 \cdot K)}{W}$$

$$q''_2 = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{th,2}} = \frac{1616 - 400}{0.0038} = 320 \frac{KW}{m^2}$$

$$T_o - T_{\infty,i} = q''_2 \Rightarrow T_o = T_{\infty,i} + q''_2 (R''_{cond,b} + R''_{conv,i})$$

$$T_o = T_{\infty,i} + q''_2 \left(\frac{L_b}{K_b} + \frac{1}{h_i} \right) = 400 + (320 \times 10^3) \left(\frac{(5 \times 10^{-3})}{25} + \frac{1}{500} \right)$$

$T_o = 1104 K \rightarrow$ Superalloy blade maintained under T_{max} ($1104 < 1200 K$)